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# Fold-Change Estimation of Differentially Expressed Genes using Mixture Mixed-Model

Arief Gusnanto, Alexander Ploner, and Yudi Pawitan

## Abstract

Microarray experiments produce expression measurements for thousands of genes simultaneously, though usually for a small number of RNA samples. The most common problem is the identification of genes that are differentially expressed between different groups of samples or biological conditions. As the number of genes far exceeds the number of RNA samples, the inherent multiplicity poses a severe problem in both hypothesis testing and effect estimation. While much of the recent literature is focused on the hypothesis aspects, we concentrate in this paper on effect estimation as a tool for the identification of differentially expressed genes. We propose a linear mixed model where the random effects are assumed to follow a mixture distribution, and study in detail the case of three normals, corresponding to genes that are down-, up- or non regulated. Our approach leads to a new type of non-linear shrinkage estimation, where a proportion of estimates is shrunk to zero, while the rest follows standard linear shrinkage. This allows us to estimate the log fold-change of the genes involved and to identify those that are differentially expressed within the same model framework. We investigate the operating characteristics of our method using simulation and spike-in studies, and illustrate its application to real data using a breast-cancer dataset.

**KEYWORDS:** mixture distribution, linear mixed model, microarrays, differentially expressed

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# 1 Introduction

Microarray technology enables us to obtain the expression levels of thousands of genes simultaneously and to study gene regulation at the RNA level. One of the most common problems in analysing this kind of expression data is the identification of genes that are differentially expressed (DE) between biological conditions or sample groups. Most of the available literature is focused on the multiplicity problem in hypothesis testing. In this paper, we study the corresponding estimation of differential expression, or fold changes. Specifically, we investigate a mixed model where the random effects – representing the log fold-changes – are assumed to come from a mixture model.

The most popular current approach to dealing with multiplicity is control of the false discovery rate as proposed by Benjamini and Hochberg (1995). This concept has inspired a vast literature, from which we will only select a few key papers relevant to mixture models in order to establish the background for our approach.

A nonparametric empirical Bayes method was proposed by Efron *et al.* (2001), based on the assumption that the distribution of (modified)  $t$ -statistics is a mixture of test statistics for DE and nonDE genes. Other applications of mixture models include Allison *et al.* (2003), who investigate the distribution of gene-wise p-values derived from a generic test statistic; their null hypothesis that genes are not affected by treatment corresponds to a uniform distribution of the p-values, while the alternative hypothesis assumes the p-values are distributed as a mixture of uniform and beta distributions. Genes of interest are selected by setting an a priori threshold for the p-values.

Kauermann and Eilers (2004) study a mixed model where the random effects are assumed to follow a mixture of zeros (zero effect) and a normal distribution  $N(0, \sigma_b^2)$ ; the zeros correspond to the genes that are not DE, and both up- and down-regulated genes are modelled by the single normal distribution. Newton *et al.* (2004) propose a hierarchical mixture model: on the first level, the model assumes that gene expression values (in the original scale) follow a gamma distribution for each of the conditions; on the second level, the distribution of the expected gene expression values is assumed to follow a mixture distribution of three hypotheses: equivalent expression, under-expression under the first condition, and over-expression under the first condition. Other methods involving mixture models were proposed by Ghosh (2004) and Irizarry *et al.* (2003b).

Additionally to identifying DE genes, researchers are often interested in estimating expression fold changes. As for the identification of DE genes, multiplicity presents a problem for estimation. Procedures aimed at addressing this problem have been proposed by Kerr *et al.* (2002) within their ANOVA model, Newton *et al.* (2004) within their hierarchical model, and Kauermann and Eilers (2004) within their threshold mixture model.

In this paper, we investigate a general linear mixed model for gene expression

data, where the random effects are assumed to follow a finite mixture distribution. We present an application of the model to a real breast cancer dataset, comparing estrogen receptor (ER)-positive and ER-negative patients. We also validate the method in a spike-in study, showing that we can correctly identify the spike-in genes. Finally, a simulation study is performed to check the model implementation, and to obtain the operating characteristics of the mixture mixed-model under two different variance assumptions.

## 2 Statistical methodology

### 2.1 Mixture mixed-model

In this section, we present the model for the general case; the specific application to gene expression data is presented in Section 2.3. The mixture mixed-model specifies

$$\begin{aligned} y &= X\beta + Zb + \varepsilon \\ \varepsilon &\sim N(0, \Sigma) \\ b &\sim \sum_{j=1}^g \pi_j N(\mu_j, D_j), \quad D_j = \sigma_{bj}^2 I_q \\ \sum_{j=1}^g \pi_j &= 1 \\ 0 &\leq \pi_j \leq 1, \quad j = 1, \dots, g \end{aligned} \tag{1}$$

where  $y$  is an  $n$ -vector of responses,  $X$  is the design matrix of size  $n \times p$  to the fixed effect vector  $\beta$  of length  $p$ ,  $Z$  is the design matrix of size  $n \times q$  of the random effect vector  $b$  of length  $q$ ,  $\varepsilon$  is the  $n$ -vector of error terms, assumed to follow a normal distribution with zero mean and variance-covariance matrix  $\Sigma = \sigma^2 I_n$ , where  $I_n$  is an identity matrix of size  $n$ , and  $\pi_1, \dots, \pi_g$  are the mixing proportion or weights.  $b$  and  $\varepsilon$  are assumed to be independent. In the ordinary normal linear mixed model,  $b$  is typically assumed to follow a normal distribution with mean  $\mu \equiv 0$  and variance-covariance matrix  $D = \sigma_b^2 I_q$ . This assumption is modified in the mixture mixed model, where we assume that  $b$  follows a mixture of  $g$  normal distributions.

For the application of Model (1) to microarray data Section 2.3, we will define  $y$  as the log fold-change of gene expression,  $X\beta$  as a fixed intercept,  $Z$  as the identity matrix, and  $b$  as the gene effects.

The formulation of Model (1) is more general than those of Newton *et al.* (2004) and Kauermann and Eilers (2004). In Newton *et al.* (2004), the comparison is only between two conditions, while Model (1) allows comparisons for any number of conditions. A study of Ishwaran and Rao (2003) presents a modification of generalized ridge regression that can be adapted to the comparison of more than two groups. The assumption of the parameters of gene effects in the model can be regarded as a special case of the random effects assumption in Model (1), with a mixture of two normals with zero mean and different variances. A study by Cui *et al.* (2005) can also be adapted to the comparison of multiple groups, but DE genes are identified through modeling the variance components in linear models. As will be described

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in Section 2.3), in the mixture mixed-model DE genes are identified based on the estimated fold-changes.

In Kauermann and Eilers (2004), DE genes are assumed to follow a single normal distribution with zero mean, while non-DE genes are assumed to have zero effect. This a special case of Model (1), when all  $\mu_{js}$  are set to zero, or when there are only  $g = 2$  mixture components. For longitudinal studies, Verbeke and Lesaffre (1996) investigate the impact of the normality assumption for the random effects in the linear mixed model. Since the purpose of the mixture distribution assumption is to reflect the heterogeneity of the random effects population, they refer to the model as the heterogeneity model. In their model, the random effects appear as subject specific parameters, so their results do not apply in our case.

For experiments producing gene expression data, we generally expect that some genes will be down-, up-, or non-regulated between conditions. Consequently, we assume that the random effects in (1) follow a mixture of  $g = 3$  normal distributions. For convenience, we will therefore only consider the situation  $g = 3$  in detail, although the methodology applies generally. An objective choice of the number of components can be based e.g. on the Akaike information criterion.

## 2.2 Parameter estimation

Parameter estimation is based on the hierarchical likelihood (Lee and Nelder, 1996) or extended likelihood (Pawitan, 2001) approach. Defining  $\theta \equiv \{\mu_j, D_j, \pi_j, \Sigma = \sigma^2 I_n, j = 1, \dots, g\}$ , the extended likelihood can be written as

$$L(\beta, \theta, b) = p(y|b)p(b). \quad (2)$$

The mixture mixed-model (1) can be expressed as

$$y_k = x'_k \beta + z'_k b + \varepsilon_k \quad (3)$$

for  $k = 1, \dots, n$ , where  $y = \{y_k\}$ ,  $x'_k$  is a  $p$ -vector, and  $z'_k$  is a  $q$ -vector. The log (extended) likelihood for each  $k$  can be written as

$$\begin{aligned} \log L_k(\beta, \theta, b) &= -\frac{1}{2} \log \sigma^2 - \frac{1}{2\sigma^2} (y_k - x'_k \beta - z'_k b)^2 \\ &\quad + \sum_{j=1}^g \log \left( \sum_{j=1}^g \pi_j p_j(b_i) \right). \end{aligned} \quad (4)$$

Taking the derivative of  $\log L_k(\beta, \theta, b)$  with respect to  $b$  gives us

$$\begin{aligned} \frac{\partial}{\partial b} \log L_k(\beta, \theta, b) &= \frac{1}{\sigma^2} z_k (y - x'_k \beta - z'_k b) + \left\{ \frac{\sum_{j=1}^g \pi_j \frac{\partial p_j(b_i)}{\partial b_i}}{\sum_{j=1}^g \pi_j p_j(b_i)} \right\} \text{ for } i=1, \dots, q \\ &= \frac{1}{\sigma^2} z_k (y - x'_k \beta - z'_k b) \\ &\quad + \left\{ \frac{\sum_{j=1}^g \pi_j p_j(b_i) \frac{b_i - \mu_j}{\sigma_{b_j}^2}}{\sum_{j=1}^g \pi_j p_j(b_i)} \right\} \text{ for } i=1, \dots, q \end{aligned} \quad (5)$$

We define

$$p_{ij} = \frac{\pi_j p_j(b_i)}{\sum_l \pi_l p_l(b_i)}, \quad (6)$$

and  $P_j$  as a diagonal matrix of size  $q \times q$  with diagonal entries  $p_{ij}$  for  $i = 1, \dots, q$ . The expression (5) can be written as

$$\begin{aligned} \frac{\partial}{\partial b} \log L_k(\beta, \theta, b) = & - \left( \frac{1}{\sigma^2} z_k z'_k + \text{diag}_i \left\{ \sum_{j=1}^g p_{ij} \frac{1}{\sigma_{bj}^2} \right\} \right) b \\ & + \left( \frac{1}{\sigma^2} z_k (y_k - x'_k \beta) + \text{vector}_i \left( \sum_{j=1}^g p_{ij} \frac{\mu_j}{\sigma_{bj}^2} \right) \right). \end{aligned} \quad (7)$$

Combining for  $k = 1, \dots, n$ , we arrive at

$$\begin{aligned} \frac{\partial}{\partial b} \log L(\beta, \theta, b) = & -(Z' \Sigma^{-1} Z + \sum_{j=1}^g P_j D_j^{-1}) b \\ & + (Z' \Sigma^{-1} (y - X \beta) + \sum_{j=1}^g P_j D_j^{-1} \mu_j), \end{aligned} \quad (8)$$

with  $\Sigma = \sigma^2 I_n$  and  $D_j = \sigma_{bj}^2 I_q$ . Equating the above expression to zero yields the random effects estimate

$$\begin{aligned} \hat{b} = & (Z' \Sigma^{-1} Z + \sum_{j=1}^g P_j D_j^{-1})^{-1} Z' \Sigma^{-1} (y - X \beta) \\ & + (Z' \Sigma^{-1} Z + \sum_{j=1}^g P_j D_j^{-1})^{-1} (\sum_{j=1}^g P_j D_j^{-1} \mu_j). \end{aligned} \quad (9)$$

The estimate of the fixed effects  $\beta$  can be obtained in a similar manner, and can be written as

$$\hat{\beta} = (X' \Sigma^{-1} X)^{-1} X' \Sigma^{-1} (y - Z \hat{b}). \quad (10)$$

Important parameters in the formulation of the extended likelihood in (2) are the mixing proportions or weights  $\pi_j$ , for  $j = 1, \dots, g$ . The estimation of  $\pi_j$  can be derived from the adjusted profile likelihood (Lee and Nelder, 1996). Interestingly, the previous algorithm can be re-expressed in terms of an expectation-maximization (EM)-like algorithm. The advantage of the EM-approach is that the quantities more easily recognizable, and it yields an explicit objective function for estimating the  $\pi_j$ 's. Note, however, that this is not a standard EM algorithm, especially in the expectation step (E-step), which involves two random quantities: random effects and group indicators. In the expectation step, we take the expectation of the group indicator with respect to the observed data and given the random effects. The random effects are then estimated in the maximization step (M-step). We will show that this yields the same random effects estimates as the extended likelihood. We also obtain the same 'weights'  $p_{ij}$ , that can be then interpreted as the probability of each gene belonging to one of the mixture components.

We first introduce  $\delta = (\delta_1, \dots, \delta_q)$ , where  $\delta_i$  is an indicator variable for  $b_i$  to belong to one of the three mixture components, with  $P(\delta_i = j) = \pi_j$ . Based on  $\delta$ , we can formulate the problem of estimating the parameters of Model (1) as an incomplete data problem: if we knew all  $\delta_i$ , then the estimation in Model (1) can

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be done in a straightforward manner. Defining  $\theta \equiv \{\mu_j, D_j, \pi_j, \Sigma, j = 1, \dots, g\}$ , the complete likelihood can be written as

$$L_c(\beta, \theta, b, \delta) = p(y|b, \delta)p(b|\delta)p(\delta) \quad (11)$$

where

$$\begin{aligned} p(y|b, \delta) &= p(y|b) = \phi(X\beta + Zb, \Sigma) \\ p(b|\delta = j) &= \phi(\mu_j, D_j). \end{aligned} \quad (12)$$

$\phi$  in (12) represents the normal probability density function, and we can see that the dependence on  $\delta$  in  $p(y|b, \delta)$  is redundant. In other words, we can construct  $y$  from  $b$  alone, and we can rewrite (11) as

$$L_c(\beta, \theta, b, \delta) = p(y|b)p(b|\delta)p(\delta). \quad (13)$$

Based on this formulation, we define the E-step: starting with initial values  $\beta^0, b^0$ , and  $\theta^0$ , we compute the expected value of the complete log likelihood conditional on the observed value

$$\begin{aligned} Q(\beta, \theta, b) &= Q(\beta, \theta, b|\beta^0, \theta^0, b^0) \\ &= E\{\log L_c(\beta, \theta, b)|y, \beta^0, \theta^0, b^0\} \\ &= E\{\log p(y|\beta^0, \theta^0, b^0) \\ &\quad + \sum_{i=1}^q \sum_{j=1}^g [I(\delta_i = j) \log p_j(b_i) + I(\delta_i = j) \log \pi_j]|y, \beta^0, \theta^0, b^0\} \\ &= \log p(y|\beta^0, \theta^0, b^0) \\ &\quad + \sum_{i=1}^q \sum_{j=1}^g [p_{ij}(\theta) \log p_j(b_i) + p_{ij}(\theta) \log \pi_j], \end{aligned} \quad (14)$$

where  $I(\cdot)$  is the indicator function and

$$p_{ij}(\theta) = E(I(\delta_i = j)|\theta^0, b^0) = \frac{\pi_j^0 p_j(b_i^0)}{\sum_k \pi_k^0 p_k(b_i^0)}, \text{ for } i = 1, \dots, q. \quad (15)$$

We can see that, using the starting values, the E-step consists of finding the conditional probabilities  $p_{ij}(\theta)$ . We are directly interested in these quantities, because  $p_{ij}(\theta)$  is the conditional probability that gene  $i$  belongs to group  $j$ , and we would like to use the for classifying each gene. The expression  $p_{ij}(\theta)$  in (15) is the same as the weight  $p_{ij}$  in (6).

In the M-step, we maximize

$$Q(\beta, \theta, b) = \log p(y | \beta^0, \theta^0, b^0) + \sum_{i=1}^q \sum_{j=1}^g [p_{ij}(\theta) \log p_j(b_i) + p_{ij}(\theta) \log \pi_j] \quad (16)$$

to get update values  $\beta^1, \theta^1$  and  $b^1$ . We take the first derivative of  $Q(\beta, \theta, b)$  with respect to  $\beta, \theta$ , or  $b$ , and set the expression to zero. The derivatives are straightforward and yield closed-form formulas for obtaining the updates. For identifiability

purposes, in the case of  $g = 3$ , we associate  $j = 1$  with the population of down-regulated genes,  $j = 2$  with non-regulated, and  $j = 3$  with up-regulated genes. We also constrain the means of the mixture components so that  $\mu_1 \leq 0$ ,  $\mu_2 = 0$ , and  $\mu_3 \geq 0$ .

Let  $P_j$  be a diagonal matrix of size  $(q \times q)$  where the diagonal entries are  $p_{ij}(\theta)$  for  $i = 1, \dots, q$ . It can be shown that the update expression for  $b$  in the M-step can be written as

$$b^1 = (Z'\Sigma^{-1}Z + \sum_{j=1}^3 P_j D_j^{-1})^{-1} Z'\Sigma^{-1}(y - X\beta) + (Z'\Sigma^{-1}Z + \sum_{j=1}^3 P_j D_j^{-1})^{-1} (\sum_{j=1}^3 P_j D_j^{-1} \mu_j), \quad (17)$$

which is the same as the random effect estimate (9) that we obtain from the derivative of the extended likelihood.

## 2.3 Two-sample comparisons

We now discuss the important special case of two-sample comparisons, where we can clearly see the shrinkage effect on the estimate. Here we describe the application to single-color microarray data, though the method can easily be modified for two-color data. Let  $W$  be a matrix of size  $n \times q$  where  $w_{ki}$  is the log expression of gene  $i$  in sample  $k$ . We write log fold-change for gene  $i$  as

$$y_i = \bar{w}_{1i} - \bar{w}_{2i},$$

where  $\bar{w}_{1i} = \sum_{k \in \text{group } 1} w_{ki}/n_1$ ,  $\bar{w}_{2i} = \sum_{k \in \text{group } 2} w_{ki}/n_2$ , and  $n_1$  and  $n_2$  are the number of samples in the two groups, which can be control vs. treatment effects, cancer vs. normal tissue, or any other meaningful distinction between two different conditions. (In the case of two-color data,  $y_i$  would be the mean log ratio for gene  $i$ .)

The mixture mixed-model (1) reduces then to

$$y_i = u + b_i + e_i, \quad (18)$$

where  $y_i$  is the log fold change of gene  $i$ ,  $u$  is a fixed intercept,  $b_i$  is the random effect assumed to come from a mixture of three normal distributions, and  $e_i$  is the error term and independent of  $b_i$ . Note that the error term can be specified in two ways,  $e_i \sim N(0, \sigma^2)$ , or  $e_i \sim N(0, \sigma_i^2)$ . In the former, we assume that the error variance is common for all genes, while in the latter, we allow the error variances to be specific for each gene. In practice, we will use for both  $\sigma^2$  or  $\sigma_i^2$  the variance of  $y_i$  computed from the data. In (18), we assume implicitly that the gene expressions have been properly normalized, so that the array effect is not included in the model.

At convergence of the iterations, the random effects estimates of model (18) assuming common error variance and  $u = 0$  can be seen as

$$\hat{b}_i = \left( \frac{1}{1 + k_i \sigma^2} \right) y_i + \left( \frac{\sigma^2}{1 + k_i \sigma^2} \right) \sum_j v_{ij} \mu_j, \quad (19)$$

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where

$$v_{ij} = \frac{p_{ij}(\theta)}{\sigma_{bj}^2}$$

and  $k_i = \sum_j v_{ij}$ . As an illustration how the estimates are related to  $y$ , let us assume that  $\sigma_{bj}^2 = \sigma_b^2$  for all  $j = 1, \dots, g$ . We can rewrite the estimates in (19) as

$$\hat{b}_i = \left( \frac{\sigma_b^2}{\sigma_b^2 + \sigma^2} \right) y_i + \left( \frac{\sigma^2}{\sigma_b^2 + \sigma^2} \right) \sum_j p_{ij}(\theta) \mu_j. \quad (20)$$

Under the individual error variance assumption, the estimates in (19) can be written as

$$\hat{b}_i = \left( \frac{1}{1 + k_i \sigma_i^2} \right) y_i + \left( \frac{\sigma_i^2}{1 + k_i \sigma_i^2} \right) \sum_j v_{ij} \mu_j. \quad (21)$$

Equation (20) indicates that at fixed  $\sigma_b^2$ , when the error variance is large, the contribution of  $y_i$  to the estimates becomes small, and the estimates will recover to  $\mu_i$ . At fixed  $\sigma^2$ , when  $\sigma_b^2$  is large, the random effects estimates tend toward  $y_i$ , and the contribution of  $\mu_i$  becomes smaller.

To illustrate the relation of  $\hat{b}$  to the raw log fold-change  $y$ , Figure 1(a) shows  $\hat{b}$  as a function of  $y$ , assuming common error variance based on (19) at a given value of  $\pi_j, \mu_j, \sigma_{bj}^2, \sigma^2$  and  $g = 3$ . The estimates are shrunk towards each of the  $\mu_j$ 's, but by different amounts. In particular, values near zero are shrunk severely toward zero, but values away from zero are only weakly shrunk. As a comparison, Figure 1(b) shows the estimates  $\hat{b}$  under the standard linear mixed-model, where the random effects  $b$  are assumed to follow a single normal distribution. The figure shows that all of the log fold-changes are shrunk toward zero by the same factor. It is clear that in microarray applications, the shrinkage effect in (a) is more appealing than in (b).

Figure 1(a) shows some interesting similarities with other nonlinear shrinkage methods in the literature. Figure 1(c) shows the so-called hard thresholding (Fan and Li 2001)

$$\hat{b} = yI(|y| > \lambda),$$

where  $\lambda = 0.5$ , and  $I(\cdot)$  is an indicator function equal to one if the expression inside the parentheses is true and zero otherwise. Figure 1(d) shows the so-called soft thresholding (Fan and Li 2001) or shrunk centroids (Tibshirani *et al.* 2002)

$$\hat{b} = \text{sign}(y)(|y| - \Delta)_+,$$

with  $\Delta = 0.5$ . In (a), (c), and (d), where genes with observed log fold-change close to zero are shrunk toward zero. Other genes with bigger observed log fold-changes in absolute value are shrunk in (a) and (d), but kept the same in (c).

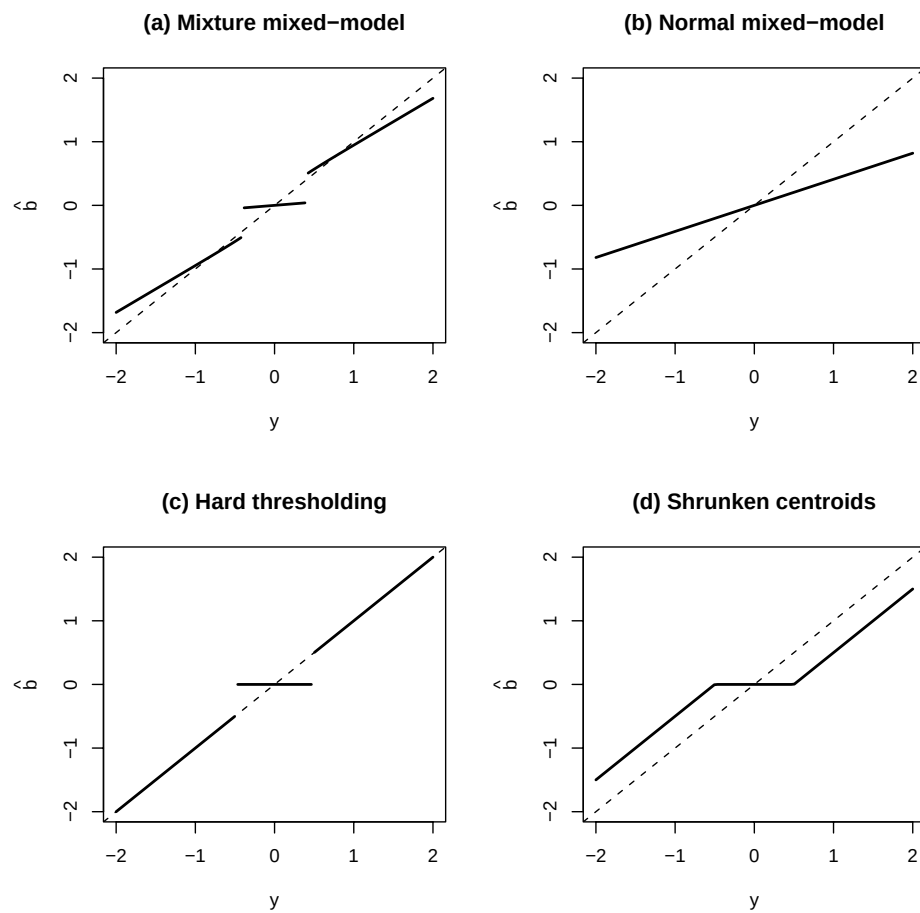


Figure 1: Illustration of  $\hat{b}$  as a function of  $y$  in several situations: (a) under the mixture mixed-model assuming common error variance where  $\sigma^2 = 0.3^2$ ,  $\sigma_{b1}^2 = \sigma_{b3}^2 = 0.5^2$ ,  $\sigma_{b2}^2 = 0.1^2$ ,  $\pi_1 = 0.1$ ,  $\pi_2 = 0.8$ ,  $\pi_3 = 0.1$ ,  $\mu_1 = -0.8$ ,  $\mu_2 = 0$ ,  $\mu_3 = 0.8$ , (b) under the (single) normal linear mixed-model with  $\mu \equiv 0$ ,  $\sigma^2 = 0.3^2$  and  $\sigma_b^2 = 0.25^2$ , (c) using hard thresholding with  $\lambda = 0.5$ , and (d) soft thresholding or shrunk centroids with  $\Delta = 0.5$ . All figures are compared to the dashed identity line.

## 2.4 Inference

The inference on gene  $i$  is based on the posterior probability that it belongs to the  $j$ th mixture component  $p_{ij}(\theta)$  provided by the EM algorithm as stated in (15). We infer that gene  $i$  belongs to a particular mixture component, if its posterior probability for that component is higher than for the other components. With this type of inference, we can identify the genes that belong to the mixture components corresponding to DE genes. From Figure 1(a), where  $g = 3$ , we can see that the list of DE genes contains genes that have random effects estimates different from zero.

Researchers often use microarrays in their experiments to obtain a short list of DE genes as candidates for further analysis with more sophisticated methods, such as real-time polymerase chain reaction (PCR). To obtain this short list of DE genes, we can first order the DE genes based on  $p_{ij}(\theta)$  from the highest to the lowest, where we fix  $j = 1, 3$ . For the next step, we cut the list at some point chosen *a priori*, depending for example on whether we are interested in a fixed number of top genes, or whether we want to control the percentage of false positives. In the second case, we select genes that have posterior probability  $p_{ij}(\theta)$  above a threshold  $c$ , so that

$$P(p_{ij}(\theta) > c | \text{all genes are non-regulated}) < \alpha, \quad (22)$$

for at least one gene, where  $\alpha$  is set for instance at 0.05 or 0.1. This is in accordance with the formulation of Kauermann and Eilers (2004) for controlling false positives. We consider  $p_{ij}(\theta)$  as probability in a multiple testing problem, and apply suitable multiplicity corrections. As an illustration, we consider the Bonferroni correction and Holm's procedure (see Dudoit *et al.* 2002), although other correction methods also apply. Using the Bonferroni method, we can set  $c = 1 - \alpha/q$ ; for the Holm procedure, we set  $c = \min(p_{ij}(\theta) | \tilde{p}_i < \alpha)$ , where  $\tilde{p}_i = 1 - p_{ij}(\theta)$  is the adjusted p-values.

## 2.5 Standard error

Standard errors of  $\pi_j$  and  $\mu_j$ , for  $j = 1, 2, 3$ , and  $b_i$  for  $i = 1, \dots, q$ , are obtained from the observed information  $I(\beta, \theta, b|y)$ . Defining  $\Psi = (\beta, \theta, b)$ , the observed information can be expressed in terms of the derivative of the E-step function  $Q(\Psi|\Psi^0)$  (Pawitan, 2001, Ch. 12.9)

$$I(\Psi^0|y) = -\left\{ \frac{\partial^2 Q(\Psi|\Psi^0)}{\partial \Psi^2} + \frac{\partial^2 Q(\Psi|\Psi^0)}{\partial \Psi \partial \Psi^0} \right\} \Big|_{\Psi=\Psi^0}. \quad (23)$$

## 2.6 Other applications

We briefly describe applications of Model (18) for purposes other than identification and estimation of DE genes, in particular modelling of the variance rather than the mean, either directly to address variance heterogeneity between genes, or through the (log) ratio of variances, to address variance heterogeneity between groups.

### 2.6.1 Modelling the variance

In Model (18), the error variance  $\sigma_i^2$  can be assumed to be common to all genes or specific for each gene. We can modify this assumption so that we require some genes to have common error variance, and to allow the other genes to have specific error variances. We call this the combined error-variance assumption.

First, we need to classify the variance of the log fold-change  $y$  of Model (18) into three different groups of low, medium, and high variability. Let  $s_{1i}^2$  and  $s_{2i}^2$  be the sample variance of the first and second group in gene  $i$ , respectively. We define  $lv_i$  to be the log of the variance estimate of  $y_i$

$$lv_i = \log \left( \frac{(n_1 - 1)s_{1i}^2 + (n_2 - 1)s_{2i}^2}{n_1 + n_2 - 2} \right) \left( \frac{1}{n_1} + \frac{1}{n_2} \right).$$

Once we identify the genes in each group, we allow genes with high  $lv$  to have specific error variances in Model (18), and force genes with low and medium  $lv$  to have common error variance.

Correspondingly, the modelling of log fold-change in Model (18) is done in two stages: first we model

$$lv_i = u + b_i^l + e_i^l, \quad (24)$$

where  $u$  is the intercept,  $b^l = \{b_i^l\}, i = \dots, q$ , are the random effects estimates, assumed to follow a mixture of  $g = 3$  normal distributions as described in Model (18),  $e_i^l$  is an error term independent of  $b_i^l$ . We fit the model under the assumption of common error variance ( $e_i^l \stackrel{\text{ind}}{\sim} N(0, \sigma^{2l})$ ). In the second stage, we apply Model (18) under the combined error-variance assumption. We will demonstrate this two-stage approach for spike-in study described in the next section.

When working with real microarray data, especially when sample sizes are small, the purpose of classifying the low error-variance genes with the medium error-variance genes is to avoid false significant results that are due to exceedingly small log fold-changes. These kind of false positive are well known in the literature, and the usual way of dealing with this problem is to add an empirically derived constant to the individual variance estimates (Tusher *et al.* 2001, Efron *et al.* 2001, Tibshirani *et al.* 2001). Our combined approach offers similar protection against inflated test statistics, but within the framework of the mixture mixed-model.

### 2.6.2 Modelling the variance ratio

Consider the question whether  $\sigma_{1i}^2 = \sigma_{2i}^2$  for gene  $i$ , for instance in the two-sample t-test. We can formulate this question in our model framework by defining  $v$  as the log of the ratio between sample-variance from the two conditions

$$v_i = \log \frac{s_{1i}^2}{s_{2i}^2},$$

and model

$$v_i = u + b_i^v + e_i^v, \quad (25)$$

where  $u$  is the intercept,  $b_i^v$  is the random effect of gene  $i$ , and  $e_i^v$  the error term, with the same assumptions as for Model (18). Based on (25), we can infer whether gene  $i$  has the same variances under both conditions. The application of this model will not be discussed here.

### 2.6.3 Software

An R package for the methods in this paper is available at <http://www.meb.ki.se/~arigus/R/mixture>

## 3 Datasets

### 3.1 Breast cancer data

This dataset is from a study in Karolinska Hospital and Karolinska Institutet, Sweden, involving 159 breast cancer patients. The RNA from frozen tumor tissues of the patients were hybridized on Affymetrix high-density oligonucleotide arrays human HG-U133A chips. Scanned image files were visually inspected for artifacts and analyzed with Microarray Suite 5.0. This resulted in expression data for 22,283 probe sets and 159 breast primary tumors. Only probe sets which were detected in more than half of the tumors were included in the subsequent analysis. The detection was performed using the Affymetrix GeneChip<sup>®</sup> Expression Analysis algorithm (Absent Call; p-value > 0.06). After this step, 11333 probesets are available for further analysis.

On each patient, we have information on the estrogen receptor (ER) level; the higher the level, the more a breast cancer responds to estrogen. Based on the ER level, we categorize whether a breast cancer is ER-positive or ER-negative. The ER status is an important clinical characteristic important in determining what kind of therapy is suitable for a patient. Among our 159 patients, 130 are ER-positive and the remaining 29 are ER-negative. Our interest is to identify the DE genes common to all patients and to estimate their fold changes.

### 3.2 Spike-in dataset

This a GeneLogic dataset where the DE genes are known (Antonellis *et al.* 2002). In the experiment, 11 cloned bacterial and phagemid RNAs with known concentration were added to RNA from an acute myeloid leukemia (AML) cell line. The materials were hybridized on U95A Affymetrix microarrays. There were 97 microarrays in total involved in the experiment, but only six of them are considered in our analysis. The six arrays are from the 25 pM and 50 pM concentration, each group consisting of three arrays. In this dataset, our interest is to identify the genes that are known

to be DE in comparison between the two concentrations. The probe-level analysis is based on the robust multi-chip analysis (RMA) method (Irizarry *et al.* 2003a).

### 3.3 Simulated dataset

We perform a simulation study to assess the proposed model (18), and to compare the operating characteristics of the two basic variants (common vs. individual variance) for a small number of samples. The overall setup is similar to that described in Newton *et al.* (2004): We set the number of samples to  $n = 6$ , three in each group, and the number of genes to  $q = 5000$ . We study four different scenarios with varying degrees of differential expression, and simulate for all of them using both common and individual variances.

The basic scenarios are based on the true random effects in the simulation,  $b^*$ :

0.  $b^* \sim \pi_1^* N(-0.5, 0.05) + \pi_2^* N(0, 0) + \pi_3^* N(0.5, 0.05)$
1.  $b^* \sim \pi_1^* N(-0.5, 0.05) + \pi_2^* N(0, 0.05) + \pi_3^* N(0.5, 0.05)$
2.  $b^* \sim \pi_1^* N(-1, 0.05) + \pi_2^* N(0, 0.05) + \pi_3^* N(1, 0.05)$
3.  $b^* \sim \pi_1^* N(-2, 0.05) + \pi_2^* N(0, 0.05) + \pi_3^* N(2, 0.05)$ ,

where the  $\pi_j^*$  are the proportions of the mixture components corresponding to down-regulated ( $j = 1$ ), non-regulated ( $j = 2$ ), and up-regulated genes ( $j = 3$ ).

For each of the scenarios except Scenario 0, we use two different probability levels for differential expression: (a)  $P(DE) = 0.2$  with  $\pi_1^* = \pi_3^* = 0.1$  and  $\pi_2^* = 0.8$ , and (b)  $P(DE) = 0.1$  with  $\pi_1^* = \pi_3^* = 0.05$  and  $\pi_2^* = 0.9$ . For Scenario 0, we only use  $P(DE) = 0.2$ .

For each scenario, the simulated log fold-change is generated by

$$y_i^* = b_i^* + e_i^* \quad i = 1, \dots, 5000.$$

For each scenario, the error terms  $e_i^*$  are generated for both variants of the model: (a) common variance for all genes,  $e_i^* \sim N(0, 0.05)$ , and (b) individual variance for each gene,  $e_i^* \sim N(0, \tau_i^2)$ , where the variances are from a log-normal distribution,  $\log \tau^2 \sim N(-5, 1)$  with quantiles  $q_{0.05} = 0.001$  and  $q_{0.95} = 0.035$ .

We fit model (18) under the assumption of both common and individual error variances. The results from these model fits are shown in Table 2 and discussed in Section 4.3. We also include the results from using classical t-statistics and the false discovery rate (FDR, Benjamini and Hochberg 1995) to identify DE genes. We choose two different cutoffs for false discovery rates:  $q^* = 0.05$  and  $q^* = 0.2$ ; the rather large second cutoff value is included to study the sensitivity of this approach to detect truly DE genes.

In this simulation, the level of difficulty for identifying DE genes is set by varying the mean of the mixture components in the distribution of  $b^*$  corresponding to the

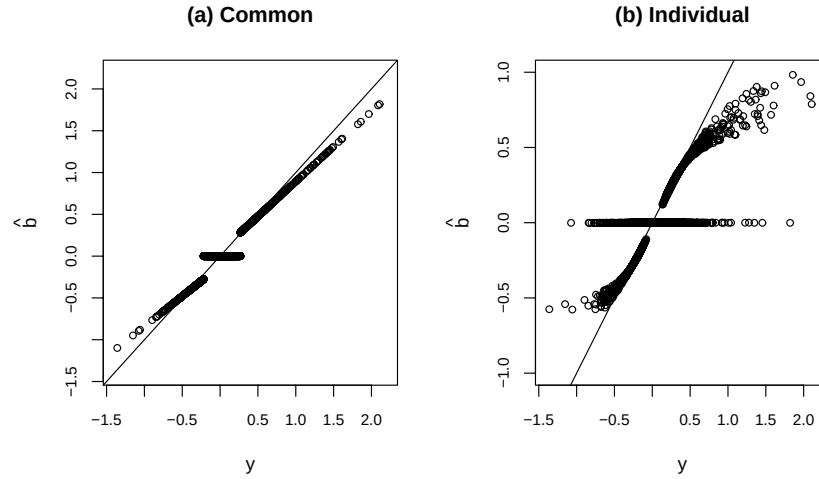


Figure 2: The estimates based on model (18) in the breast cancer dataset, assuming (a) common error variance, and (b) individual error variance, compared to identity line.

DE genes. In each of the scenarios, the variance of the mixture components is kept constant. This is equivalent to varying the variance of the mixture components and fixing the means of the component distributions.

## 4 Results

### 4.1 Breast-cancer dataset

In the breast cancer dataset, our interest is to identify the genes that are DE between ER-positive and ER-negative patients. Figure 2 (a) shows the random effects estimates  $\hat{b}$  compared to the observed log fold-change  $y$  assuming common error-variance. The figure shows that the estimates are separated into three different groups; the genes corresponding to the nonDE group have fold changes close to zero as seen in Figure 1(a). The remaining genes are classified as down- ( $\hat{\mu}_1 = -0.37$ ,  $se=0.005$ ) and up-regulated ( $\hat{\mu}_3 = 0.46$ ,  $se=0.007$ ). The estimates of proportions  $\hat{\pi}$  are 0.082 ( $se=0.0027$ ), 0.834 ( $se=0.0086$ ), and 0.084 ( $se=0.0027$ ), corresponding to down-, non-, and up-regulated genes respectively.

Figure 2(b) shows the random effects estimates  $\hat{b}$  compared to the observed log fold-change  $y$  under the individual error-variance assumption. The estimates are also separated into three groups, corresponding to down ( $\hat{\mu}_1 = -0.28$ ,  $se=0.0059$ ), up ( $\hat{\mu}_3 = 0.37$ ,  $se=0.0061$ ), and non DE genes where the estimates are close to zero. The estimates of the mixing parameter  $\hat{\pi}$  are 0.081 ( $se=0.0027$ ), 0.845 ( $se=0.0086$ ), and 0.074 ( $se=0.0026$ ), corresponding again to the down-, non-, and up-regulated

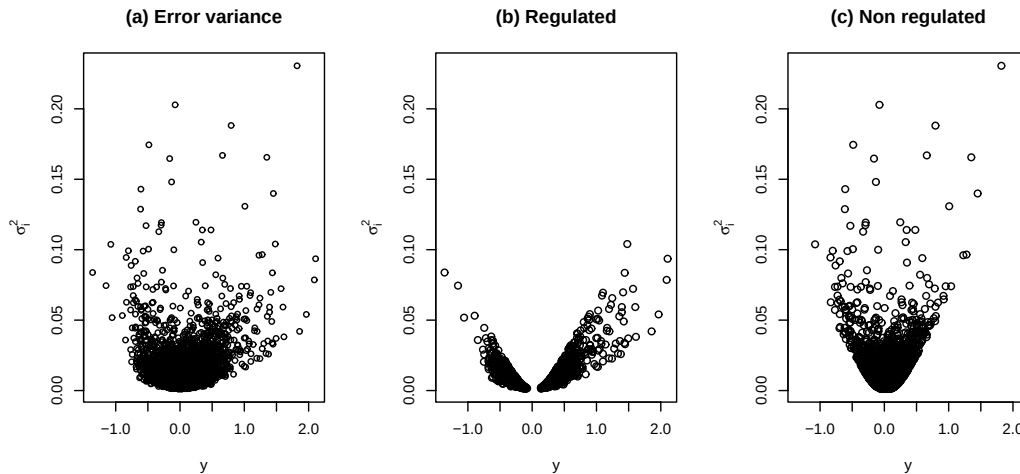


Figure 3: Comparison of  $\sigma_i^2$  and  $y_i$  in model (18) assuming individual error variance (a), and after we separate the comparison based on (b) DE and (c) non DE groups.

groups respectively. The pattern of random effects estimates however is different: Figure 2(b) shows that the shrinkage takes effect individually on each of the estimates.

It can be shown that the patterns of random effects estimates under the common and individual error-variance assumptions are significant, i.e. all random effects estimates for DE genes are significantly different from zero (with a 95 % confidence interval). For nonDE genes, the random effects estimates are not significantly different from zero.

In Figure 2, some of the genes are classified as non DE in the common error-variance model, but classified as DE in the individual error-variance model, and vice versa. The genes that are classified as non DE in Figure 2 are the ones that have high error variance  $\sigma_i^2$  relative to the log fold change  $y_i$ , see also Figure 3.

Figure 3 (a) shows the estimates of the individual error variance versus the log fold-change  $y$ . The comparison is separated for DE and non-DE group as shown in Figure 3(b) and (c) respectively. Figures 3 (a), (b) and (c) show that the genes which are identified as DE are the ones that have low error-variance relative to the log fold-change  $y$ . The genes with high log fold-change which at the same time also have high error-variances are classified as nonDE. This is different from the result under the common error-variance assumption, where the genes are classified as DE when  $y$  is high in absolute value, and as nonDE when  $y$  is close to zero.

The estimated proportion of nonDE genes based on the common error-variance assumption is  $\hat{\pi}_2 = 0.834$ , and under the individual error-variance assumption is  $\hat{\pi}_2 = 0.845$ . Storey and Tibshirani (2003) suggested to base  $\hat{\pi}_2$  on the level where the p-value distribution is constant as the p-value goes toward one. In our dataset,



this corresponds to an estimate of  $\hat{\pi}_2 = 0.55$ ; the estimate of  $\pi_2$  based on the method suggested by Efron *et al.* (2001) is 0.52. This discrepancy requires some consideration. We interpret it as being due to differences in the null hypotheses for the different approaches: under the mixed model, all DE genes have significant differential expression at a p-value of 5%. In contrast, the estimated  $\pi_2 = 0.55$  corresponds to a *sharp* null hypothesis of nonDE for all genes; the estimate from the mixed model corresponds to a wider null hypothesis that includes genes with small effects.

Efron (2004) discusses the impact of these genes of small effect on inference, and argues that the ultimate goal is the identification a subset of 'interesting' genes rather than simply 'significant' ones. Following Efron's (2004) approach, we fit a normal mixture model to the observed t-statistics:

$$F(x) = \sum_{i=1}^9 p_i N(\mu_i, 1).$$

We find the following estimates:

$$\begin{aligned}\hat{p} &= (0.02 \ 0.05 \ 0.08 \ 0.08 \ 0.47 \ 0.09 \ 0.11 \ 0.08 \ 0.03) \\ \hat{\mu} &= (-3.17 \ -1.88 \ -1.63 \ -1.40 \ -0.08 \ 1.50 \ 1.79 \ 2.15 \ 4.22)\end{aligned}$$

In the standard normal scale, the middle five components have non-significant means; if we define the genes associated with these components as 'uninteresting', we get a total proportion of  $0.08+0.08+0.47+0.09+0.11=0.83$  of uninteresting genes, very close to our estimate of  $\pi_2$ . Arguably, our procedure therefore provides some automatic restriction of the DE genes to 'interesting' ones.

## 4.2 Spike-in study

In the spike-in study, our interest is in the spiked-in genes that we know to be DE between two the concentration groups. The results are presented in Table 1. We can see that the log fold-changes of the spike-in genes are close to one (in absolute value), as we would expect from the experiment. The spike-in genes are the top ten genes with the highest log fold-changes. Compared to that, the standard t-test performs very poorly: using the associated p-values, the ranking of the spiked-in genes slips towards the middle of the gene list, or in other words, the top-gene list based on the t-test is populated by false positives.

In Table 1, the common error-variance model correctly identifies the spiked-in genes and ranks them as top-DE genes. The estimates of the log fold-change in the common model are slightly shrunk, as indicated in Figure 4(a). These shrunk estimates seem realistic, as in most real data situations, extreme observed fold-changes tend to have have relatively large variances.

The individual error-variance model in Table 1 correctly identifies the spiked-in genes as DE, but the rank of the posterior probability of eight out of ten spiked-in

| Probesets       | Fold   | Rank  | p-value | Rank     | Common | Prob. | Rank |
|-----------------|--------|-------|---------|----------|--------|-------|------|
| AFFX-BioB-M_at  | -1.211 | 1     | 0.010   | 52       | -0.966 | 1     | 1    |
| AFFX-BioC-3_at  | -0.913 | 2     | 0.010   | 51       | -0.729 | 1     | 2    |
| AFFX-BioDn-3_at | -0.912 | 3     | 0.013   | 77       | -0.728 | 1     | 3    |
| AFFX-DapX-5_at  | -0.877 | 4     | 0.006   | 36       | -0.700 | 1     | 4    |
| AFFX-BioB-5_at  | -0.871 | 5     | 0.009   | 46       | -0.695 | 1     | 5    |
| AFFX-BioC-5_at  | -0.841 | 6     | 0.016   | 101      | -0.672 | 1     | 6    |
| AFFX-BioB-3_at  | -0.824 | 7     | 0.026   | 154      | -0.658 | 1     | 7    |
| AFFX-CreX-5_at  | -0.693 | 8     | 0.047   | 333      | -0.554 | 1     | 8    |
| AFFX-DapX-M_at  | -0.665 | 9     | 0.004   | 24       | -0.531 | 1     | 9    |
| AFFX-DapX-3_at  | -0.664 | 10    | 0.015   | 94       | -0.530 | 1     | 10   |
| Probesets       | Indiv. | Prob. | Rank    | Combined | Prob.  | Rank  |      |
| AFFX-BioB-M_at  | -0.303 | 0.999 | 42      | -0.879   | 1      | 1     |      |
| AFFX-BioC-3_at  | -0.324 | 1.000 | 21      | -0.662   | 1      | 2     |      |
| AFFX-BioDn-3_at | -0.306 | 0.999 | 40      | -0.661   | 1      | 3     |      |
| AFFX-DapX-5_at  | -0.361 | 1.000 | 7       | -0.636   | 1      | 4     |      |
| AFFX-BioB-5_at  | -0.330 | 1.000 | 17      | -0.631   | 1      | 5     |      |
| AFFX-BioC-5_at  | -0.295 | 0.999 | 64      | -0.610   | 1      | 6     |      |
| AFFX-BioB-3_at  | -0.266 | 0.998 | 234     | -0.597   | 1      | 7     |      |
| AFFX-CreX-5_at  | -0.240 | 0.997 | 548     | -0.502   | 1      | 8     |      |
| AFFX-DapX-M_at  | -0.392 | 1.000 | 2       | -0.481   | 1      | 9     |      |
| AFFX-DapX-3_at  | -0.308 | 0.999 | 37      | -0.481   | 1      | 10    |      |

Table 1: *Results for the spike-in study. The probesets are the ones known to be DE. The columns in the upper part of the table are the log fold-changes and their ranks, the p-values from the standard t-test and their ranks, and the random effects estimates, the posterior probabilities, and the ranks of the probabilities based on Model (18) assuming common (Common) error variance. The columns in the lower part of the table are the random effects estimates, the posterior probabilities, and the ranks of the probabilities based on Model (18) assuming individual (Indiv.) and combined (Combined) error variances.*

genes are not at the top level. Additionally, the random effects estimates are shrunk heavily, as shown in Figure 4(b). Together this demonstrates that model with individual error variances gives results that are close to the those of the standard t-test, which are inferior for this small dataset.

For the application of Model (18) with the combined error-variance assumption, we need to classify first the  $lv$  as described in Model (24) in order to determine which genes are required to have common error-variance, and which genes are allowed to have individual error-variances. The result of this classification is shown in Figure 4(c); the  $lv$  are classified into three groups, corresponding to the small, medium, and high error variance. All spike-in genes are classified as medium variance, hence we set those genes to have common error variance in Model (18). Consequently, the mixture mixed model with the combined error assumption identifies the spiked

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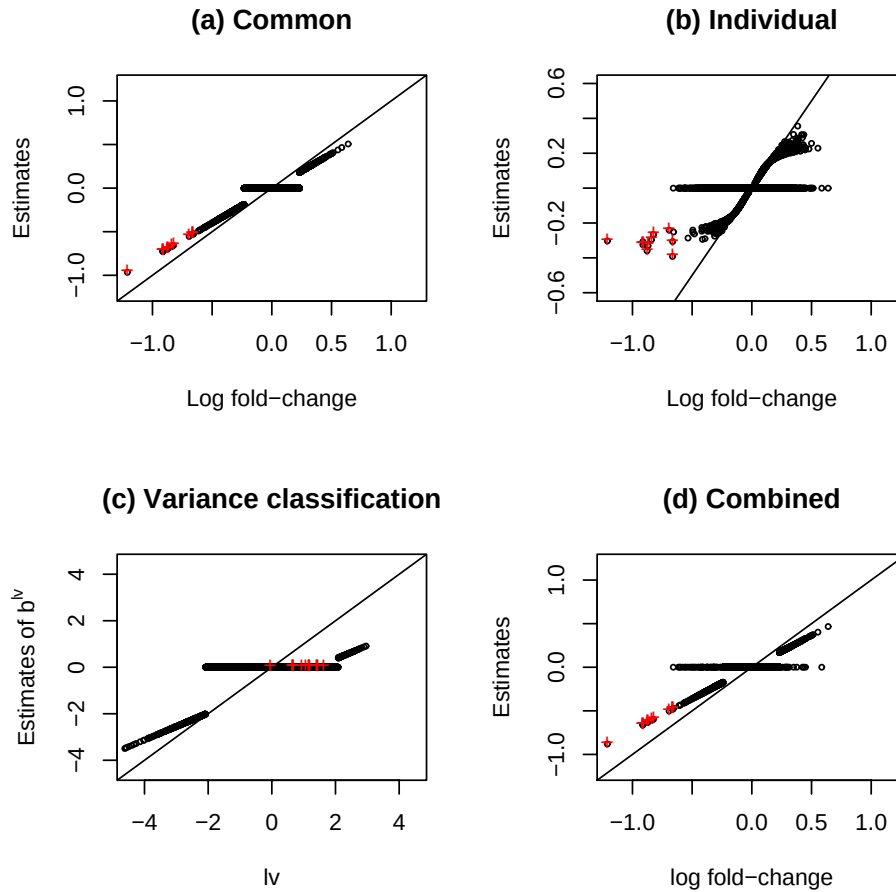


Figure 4: The estimates of the log fold-changes for the spike-in data based on Model (18) assuming (a) common and (b) individual error variances. (c) The estimates of the log variance based on Model (24) assuming common error variance. (d) The estimates of the log fold-change based on Model (18) assuming that some genes have common error variance and others individual error variances, as described in (c). In all figures, the spiked-in genes in Table 1 are marked with "+".

in genes correctly. The ranks of the posterior probabilities also list the spiked-in genes at the top. The estimates of the random effects can be seen to be shrunk in Figure 4(d), showing features of both Figure 4(a) and (b).

### 4.3 Simulation study

The result of the simulation is presented in Table 2, which lists the number of DE genes identified, realized sensitivity, specificity, false discovery rate (FDR), and

## I. Data simulated under common variance assumption

| Scenario | $P(DE)$ | Methods    | $\#DE/q$ | Sens. | Spec. | FDR  | FNDR |
|----------|---------|------------|----------|-------|-------|------|------|
| 0        | 0.2     | Common     | 0.16     | 0.58  | 0.94  | 0.28 | 0.10 |
|          |         | Individual | 0.17     | 0.50  | 0.91  | 0.41 | 0.12 |
|          |         | T05        | 0.00     | 0.00  | 1.00  | NA   | 0.20 |
|          |         | T20        | 0.00     | 0.00  | 1.00  | 0.00 | 0.20 |
| 1        | 0.1     | Common     | 0.12     | 0.45  | 0.92  | 0.61 | 0.06 |
|          |         | Individual | 0.10     | 0.34  | 0.92  | 0.66 | 0.08 |
|          |         | T05        | 0.00     | 0.00  | 1.00  | NA   | 0.10 |
|          |         | T20        | 0.00     | 0.00  | 1.00  | NA   | 0.10 |
|          | 0.2     | Common     | 0.18     | 0.48  | 0.90  | 0.46 | 0.13 |
|          |         | Individual | 0.19     | 0.40  | 0.87  | 0.57 | 0.15 |
|          |         | T05        | 0.00     | 0.00  | 1.00  | NA   | 0.20 |
|          |         | T20        | 0.00     | 0.00  | 1.00  | 0.50 | 0.20 |
| 2        | 0.1     | Common     | 0.07     | 0.62  | 1.00  | 0.05 | 0.04 |
|          |         | Individual | 0.16     | 0.34  | 0.86  | 0.77 | 0.08 |
|          |         | T05        | 0.00     | 0.00  | 1.00  | 0.00 | 0.11 |
|          |         | T20        | 0.02     | 0.17  | 1.00  | 0.20 | 0.09 |
|          | 0.2     | Common     | 0.19     | 0.89  | 0.99  | 0.06 | 0.03 |
|          |         | Individual | 0.18     | 0.42  | 0.88  | 0.53 | 0.14 |
|          |         | T05        | 0.00     | 0.00  | 1.00  | NA   | 0.20 |
|          |         | T20        | 0.08     | 0.39  | 0.99  | 0.06 | 0.13 |
| 3        | 0.1     | Common     | 0.12     | 1.00  | 0.99  | 0.07 | 0.00 |
|          |         | Individual | 0.12     | 0.68  | 0.96  | 0.34 | 0.04 |
|          |         | T05        | 0.07     | 0.60  | 1.00  | 0.04 | 0.05 |
|          |         | T20        | 0.13     | 0.98  | 0.97  | 0.18 | 0.00 |
|          | 0.2     | Common     | 0.21     | 0.96  | 0.95  | 0.05 | 0.01 |
|          |         | Individual | 0.26     | 0.99  | 0.93  | 0.23 | 0.00 |
|          |         | T05        | 0.17     | 0.82  | 0.99  | 0.04 | 0.04 |
|          |         | T20        | 0.24     | 0.99  | 0.95  | 0.17 | 0.00 |

## II. Data simulated under individual variance assumption

| Scenario | $P(DE)$ | Methods    | $\#DE/q$ | Sens. | Spec. | FDR  | FNDR |
|----------|---------|------------|----------|-------|-------|------|------|
| 2        | 0.1     | Common     | 0.09     | 0.88  | 1.00  | 0.04 | 0.01 |
|          |         | Individual | 0.09     | 0.70  | 0.98  | 0.21 | 0.03 |
|          |         | T05        | 0.00     | 0.00  | 1.00  | NA   | 0.10 |
|          |         | T20        | 0.00     | 0.01  | 1.00  | 0.09 | 0.10 |

Table 2: Results for simulated data sets: complete results for data simulated under assumption of common variance (I), only partial results for data simulated with gene-wise individual variances (II). The scenarios are those described in Section 3.3, with increasing strength of differential expression from 0 to 3.  $P(DE)$  are the different proportions of DE genes used within the scenarios. Methods shows which analysis method produced the results in a row, see below.  $\#DE/q$  is the proportion of genes selected as DE; sensitivity (Sens.) is the proportion of truly DE genes selected as DE genes; specificity (Spec.) is the proportion of truly nonDE genes selected as nonDE genes; FDR is the observed rate of false discovery in the list of DE genes, and FNDR is the observed rate of false non-discovery in the list of nonDE.

The methods for selecting DE genes are those described in Section 3.3: based on fitting model (18) assuming either common or individual error variances (Common and Individual), and based on  $t$ -tests with false discovery rate cutoff of  $q^* = 0.05$  and  $q^* = 0.2$  (T05 and T20). All results shown are the averages for two simulated datasets. NA indicates that no gene was declared DE for a particular setting.

false non discovery rate (FNDR) for different methods. Sensitivity and FDR are key quantities in planning and evaluating microarray studies (Pawitan *et al.*, 2005), but also for follow-up, since short-listed genes are often validated through other techniques like e.g. real-time polymerase chain reaction (PCR).

We find that for all simulation settings the mixture mixed-model has higher sensitivity and lower FDR with common variance (Common) than with individual variances (Individual) when the simulated log fold-changes are generated with the same variance across genes (Table 2 Part I): for example, in Scenario 2 with  $P(DE) = 0.2$ , the Common method has sensitivity 0.89 and FDR 0.06, while the Individual method has sensitivity 0.42 and FDR 0.53. As we data was generated under the common variance assumption, this is not exactly surprising.

We find the same result, however, when we use individual variances for each gene for generating the simulated log fold-changes (Table 2, Part II). For the partial results given (Scenario 2 and  $P(DE) = 0.1$ ) for example, we find that the Common method has sensitivity 0.88 and FDR 0.04, while the the Individual method has sensitivity 0.70 and FDR 0.21; the same quantitative difference can be seen for the other simulation settings (not shown). It appears that for small samples like here ( $n_1 = n_2 = 3$ ), the Common method has better performance than the Individual method, even if the assumption of a common error variance for all genes does not hold; this is in agreement with the results from the spike-in study described in the previous section.

For the methods based on the  $t$ -statistic, the sensitivity can be improved, but only at the expense of higher FDR. For example, in Scenario 3 and for  $P(DE) = 0.1$ , the sensitivity increases from 0.60 (T05 method) to 0.98 (T20 method), but the realized FDR also increases from 0.04 to 0.18.

## 5 Discussion

The standard  $t$ -test assumes that the variability of expression across samples is different between genes, which from a modelling perspective is rather wasteful. The limited amount of information on  $q$  genes in the data is used to estimate  $2q$  parameters, while we only have to estimate  $q + 1$  parameters under the assumption of common variance. When  $q$  is in the order of ten thousands like for microarray data, the difference can become important, as indicated by the results from the simulation and spike-in study. It shows that assuming common error variance, even if it is known to be unrealistic or plain incorrect, can give better results than assuming individual error variances, at least for very small sample sizes.

Kauermann and Eilers (2004) discuss the linear mixed model to identify the DE genes where the random effects follow a mixture of two distributions; the first one is a one-point distribution of zeros corresponding to the nonDE genes, the second mixture component is a normal distribution corresponding to the DE genes. An advantage of the model is that it has fewer parameters compared to the mixture

mixed model (18), but it is less flexible, especially for dealing with asymmetric regulation. In the empirical Bayes method of Efron *et al.* (2001), the empirical distribution  $f_1$  of the modified t-statistics among the DE genes suggests a mixture of normal distributions. Furthermore, the distribution of  $f_1$  is often distinct for the down- and up-regulated genes, which also makes a mixture of three normal distributions seem natural.

While sharing some properties with the model proposed by Newton *et al.* (2004), the general mixture mixed-model (18) is more flexible: e.g. it can be easily adapted to comparisons between more than two groups, whereas Newton *et al.*'s (2004) model is specifically designed for the identification of DE genes in the two-sample situation; and while an extension to more than two groups is also possible (Kendzierski *et al.* 2003), the identification of the DE genes is not based on the estimation of the log fold changes. Additionally, the random effects in (18) can be generalized for any number of mixture components  $g > 3$ , which is not possible in Newton *et al.* (2004), where the genes are classified according to one of three hypotheses. Such an extension of the number of mixture components model may be appropriate when a proportion of DE genes can be expected to be extremely strongly up- or down-regulated compared the main part of the DE genes; in this situation, it seems reasonable to attempt to fit a model with  $g = 5$ , corresponding to extremely-down regulated, medium-down regulated, non regulated, etc. genes.

The mixture mixed-model can also be used for gene classification based on other criteria than mean log fold changes, as outlined in Sections 2.6; together with the possibility of combining different variants of the mixture mixed-model, like for the combined variance assumption described in Section 2.6.1, this offers considerable flexibility in modeling gene expression data.

A possible limitation of the mixture mixed-model (18) is its fully parametric nature, and that it relies on the assumption of mixture-of-normality in the distribution of the random effects. In theory, the mixture mixed model (18) can approximate the true distribution of the random effects regardless, just by using a large enough number of components, but in practice this will require an unrealistic large number of samples. We believe that given the underlying nature of microarray technology, the number of mixture components will always be a restricted – whether this will limit the flexibility of the mixture mixed-model unduly remains to be seen.

## 6 Conclusion

We have investigated the mixture mixed model for log fold-change estimation. For a 3-component mixture, the estimates of the log fold-change are shrunk and separated into three different groups, corresponding to down-, non-, and up-regulated genes. The separation is also reflected in the posterior probability of being differentially expressed. The method is applied to a large clinical data set and validated on a spike-in study, where it identifies the spike-in genes correctly and provides shrunk

estimates the effects. In a simulation study, we find that for small samples, the assumption of a common variance across genes can lead to better results compared to using gene-wise variances, indicated by lower FDR and higher sensitivity; this holds even for situations where we know that common variance assumption does not hold.

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